

Project Initialization and Planning Phase

Date	04 July 2026
Team ID	
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform credit card application screening using machine learning, boosting efficiency and accuracy. It tackles manual bottlenecks and legacy processing delays, promising faster operations, reduced risks, and a competitive edge against FinTech competitors. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to design a comprehensive machine learning framework for automatic credit card application screening, ensuring faster, data-driven, and highly accurate risk assessments.
Scope	The project comprehensively assesses and automates the credit card approval pipeline, incorporating advanced predictive modeling to streamline customer segment validation and lower approval processing times.
Problem Statement	
Description	Traditional credit card screening relies on manual verification processes and legacy system constraints, causing major processing delays, high operational overhead, and customer frustration.
Impact	These system inefficiencies lead to severe processing bottlenecks and applicant drop-offs, making the institution lose potential market share to highly agile, real-time FinTech competitors.
Proposed Solution	
Approach	Developing a predictive automated pipeline that utilizes robust machine learning classification algorithms to analyze historical applicant data, credit history, and financial risk profiles instantly.

Key Features	- Implementation of a machine learning-based credit assessment model utilizing algorithms like XGBoost, Random Forest, or Logistic Regression.
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	- Real-time decision-making for quicker, bottleneck-free credit card approvals. - Automated feature parsing to accurately evaluate customer data segments and root out high-risk applicants early.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv